Global Environmental Issues: Climate Change, Global Warming, Impacts and Sustainable Development

TGPSC Group I | Paper IV | Unit III | Topic 5 | English | Published learning draft; NOT academically approved

Syllabus map and learning objectives

Syllabus wording (corroborated public reproduction; direct commission-hosted original not retrieved in this run): “5. Global Environment Issues: Climate Change, Global Warming and its impact, Sustainable Development.”

• Official component • Where explained • Competence •

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• Climate change • Sections 1–5, 7, 9–11 • Explain mechanisms, attribution, feedbacks and risks •

• Global warming and impacts • Sections 2–7, 12 • Interpret metrics and analyse heat, water, health, farming, ecosystems •

• Sustainable development • Sections 8–16 and applications • Evaluate equity, governance, policy trade-offs and SDGs •

Learning objectives: distinguish weather from climate and warming from climate change; explain the greenhouse effect and climate feedbacks; interpret primary-source 2026 climate facts; compare mitigation with adaptation; connect climate risks to Telangana livelihoods; evaluate global and Indian governance; and write balanced mains answers with implementable measures.

Question provenance: All 30 MCQs and all six mains prompts are original practice written for this chapter. None is represented as an official TGPSC previous-year question. Independent academic approval remains pending.

Executive summary

Global warming results primarily from human enhancement of the greenhouse effect, while climate change encompasses the wider transformation of heat, rainfall, oceans, ecosystems and extremes. The IPCC's attribution evidence is robust; uncertainties in local forecasts do not erase the global signal. Impacts are unequal because hazards interact with settlement, occupation, poverty, health and governance. Mitigation reduces emissions; adaptation reduces exposure and vulnerability; sustainable development asks whether these responses protect equity, livelihoods and ecosystems over time. Telangana's planning challenge is to combine heat-health protection, water security, resilient agriculture, urban drainage and clean energy with transparent evaluation. The Paris Agreement and India's 2026 NDC set a policy framework, but pledges must never be mistaken for achieved outcomes.

Standalone quick revision: thirty examination facts

1. Weather is short-term atmospheric condition; climate is its long-term distribution and variability.

2. Global warming is rising global mean surface temperature; climate change includes wider system shifts.

3. The enhanced greenhouse effect is caused principally by human emissions of long-lived greenhouse gases.

4. IPCC AR6: human activities unequivocally caused warming; 2011–2020 was about 1.1°C above 1850–1900.

5. WMO State of Global Climate 2025, released 23 March 2026: 2025 about 1.43°C above pre-industrial.

6. 2015–2025 were the eleven warmest years in the instrumental record.

7. Annual exceedance of 1.5°C does not alone equal a long-term Paris threshold crossing.

8. Positive feedback example: loss of reflective ice increases absorption and warming.

9. Ocean heat content integrates most of the planetary excess heat.

10. Attribution examines changes in probability and intensity, not simplistic single-event causation.
11. Hazard × exposure × vulnerability, moderated by capacity, shapes disaster risk.
12. Adaptation reduces harm from climatic effects; mitigation reduces causes of warming.
13. Maladaptation example: unlimited borewell extraction during drought depletes future groundwater.
14. Carbon dioxide intensity equals emissions divided by GDP; absolute emissions can rise as intensity falls.
15. Installed non-fossil power capacity is not the same as non-fossil electricity generation.
16. Paris Agreement: well below 2°C, pursue 1.5°C; adopted 2015, entered into force 2016.
17. CBDR-RC acknowledges differentiated responsibilities and capacities.
18. India 2035 NDC: 47% lower GDP emissions intensity than 2005 baseline.
19. India 2035 NDC: about 60% cumulative non-fossil installed electric-power capacity.
20. India 2035 NDC: 3.5–4.0 GtCO2e forest-and-tree-cover carbon sink relative to 2005.
21. India official NDC: non-fossil installed capacity 52.57% as of 28 February 2026.
22. India 2070 net-zero is a future objective, not an achieved result.
23. Telangana EPTRI: climate nodal agency designated through GO Ms No 06, 17 March 2016.
24. Telangana State Action Plan on Climate Change prepared 2015; EPTRI reports revision work.
25. Urban heat island is a local land-cover effect interacting with global warming.
26. Sustainable development balances present needs, future needs, equity and ecological limits.
27. UN 2030 Agenda: 17 SDGs, 169 targets; SDG 13 is climate action.
28. Credible forest carbon claims need additionality, permanence, leakage and verification.
29. Just transition means worker and community protection during structural decarbonisation.
30. Official PYQs require original question paper and question number; all questions here are original.

Detailed textbook notes

1. Climate system, weather, climate and attribution

Weather describes the state of the atmosphere at a place and time: temperature, humidity, precipitation, wind and pressure. Climate describes statistical distributions and variability of weather over decades, conventionally assessed using thirty-year reference periods. Climate change is a persistent shift in those distributions, including their means, variability and extremes. Global warming is the long-term increase in global mean surface temperature; it is a major component of climate change, not a synonym for all its consequences. A warmer world can produce both more intense rainfall and regional drought because circulation, soil moisture and atmospheric water-holding capacity change together. An unusually cold day does not disprove a long-term warming trend; likewise, one flood alone does not establish that climate change caused it.

The climate system couples the atmosphere, hydrosphere, cryosphere, lithosphere and biosphere. Solar radiation arrives mainly as short-wave energy, while Earth emits long-wave infrared radiation. Greenhouse gases absorb and re-emit some of this outgoing energy. The natural greenhouse effect makes Earth habitable; the enhanced greenhouse effect arises when human activity increases atmospheric greenhouse-gas concentrations. Carbon dioxide from fossil fuels and land-use change, methane from energy, livestock and waste, nitrous oxide from agriculture, and industrial fluorinated gases contribute differently because their atmospheric lifetimes, radiative efficiencies and emissions trajectories differ. Water vapour is an important feedback: warming increases the atmosphere's capacity to contain moisture, but it is not the primary externally imposed cause of modern warming.

Attribution separates natural variability from forced change. Volcanic aerosols can temporarily cool the surface; solar variation and El Niño–Southern Oscillation affect annual patterns. Yet the observed long-term global warming cannot be explained by these factors alone. The IPCC AR6 Synthesis Report (2023) states that human activities have unequivocally caused global warming and estimates 1.1°C warming for 2011–2020 relative to 1850–1900. Evidence includes thermometer records, ocean heat content, stratospheric cooling with tropospheric warming, retreating ice, isotope signatures and climate-model experiments with and without anthropogenic forcing. These independent observations form a causal argument stronger than any single indicator.

For Telangana, distinguish a long-run climate trend from a seasonal heatwave or an El Niño-related monsoon anomaly. A district heat-risk assessment must combine meteorological exposure with occupation, housing, tree cover, access to water, age and health status. A climate study should state its time period, geographical scale, reference baseline and uncertainty. In a mains answer, define the phenomenon, identify physical mechanism, present an observed indicator, discuss vulnerability, then offer an implementable response.

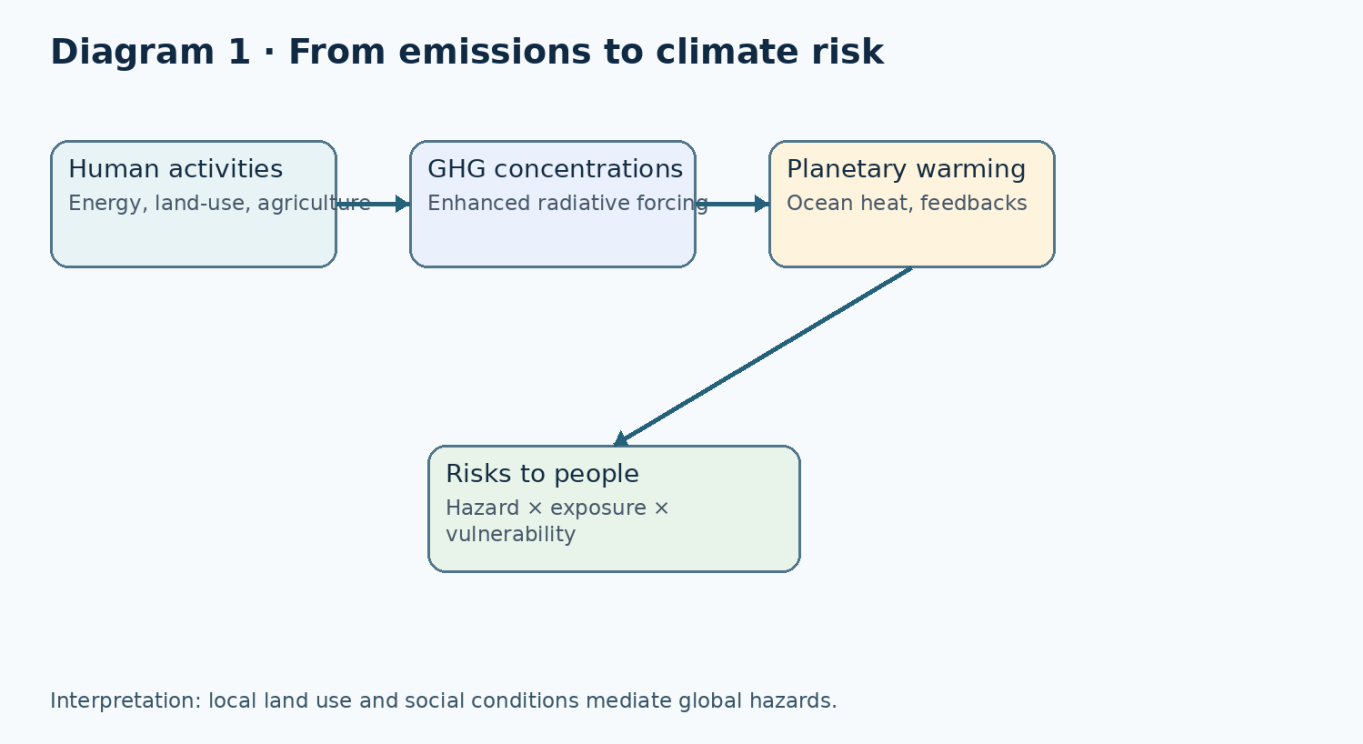
2. Earth energy balance, greenhouse gases and warming metrics

Radiative forcing is the change in Earth's energy balance associated with a driver, expressed in watts per square metre. Positive forcing tends to warm and negative forcing tends to cool. Accumulating carbon dioxide raises the concentration of a long-lived gas; the climate responds gradually because oceans absorb and release heat over long timescales. The result is an energy imbalance: more energy enters than leaves the Earth system. The World Meteorological Organization's State of the Global Climate 2025 (released 23 March 2026) reported the highest Earth energy imbalance in its 65-year record and unprecedented ocean heat content. Ocean heat is a critical indicator because oceans store most of the excess energy, even when year-to-year surface temperatures fluctuate.

Global warming potential compares the integrated radiative effect of a pulse emission of a gas with carbon dioxide over a specified horizon, often 100 years. It is a policy accounting metric, not a claim that gases have identical atmospheric lifetimes. Methane cuts can reduce near-term warming more quickly than equivalent long-lived carbon dioxide cuts, while sustained carbon dioxide reduction remains indispensable for stabilising long-term temperature. Carbon dioxide equivalent values therefore require an explicit accounting method. Carbon intensity of GDP equals emissions divided by economic output; it may fall while total emissions still rise if output grows faster than intensity falls. Non-fossil installed capacity is not the same as actual non-fossil electricity generation, and neither automatically measures economy-wide emissions.

A temperature anomaly is measured relative to a baseline. The pre-industrial benchmark usually refers to 1850–1900, while Indian weather bulletins may use another climatological normal. WMO estimated that 2025 was around 1.43°C above 1850–1900 and one of the two or three warmest years, depending on dataset. The 2015–2025 period comprises the eleven warmest years on record. Copernicus reported that 2024 was the warmest calendar year and 2025 the third warmest in its dataset. Apparent ranking differences arise from different datasets, coverage and uncertainties; they do not undermine the long-term trend.

The Paris Agreement's 1.5°C temperature objective concerns sustained long-term warming, not a single year crossing 1.5°C. A three-year or annual exceedance is an alarming signal but not a formal determination that the long-term threshold has been permanently breached. Good exam practice labels observed temperature, target, forecast and uncertainty separately. An emissions-intensity target must never be described as an absolute emissions cap; an installed-capacity target must not be reported as the share of electricity actually supplied.



Original diagram 1: 01 climate causal chain.png.

3. Feedbacks, thresholds, tipping risks and uncertainty

Climate feedbacks amplify or damp an initial forcing. Ice–albedo feedback is positive: as reflective snow and sea ice decline, darker land or water absorbs more sunlight and causes additional warming. Water-vapour feedback is positive because warmer air generally holds more moisture, itself a greenhouse gas. Some cloud responses are uncertain in magnitude and vary with altitude, type and geography. Carbon-cycle feedbacks matter when drought, wildfire or permafrost thaw release additional greenhouse gases or weaken land sinks. Feedback is not the same as an external forcing; confusing them obscures causal analysis.

A tipping element is a component of the Earth system that may shift to a different state after a threshold is crossed, sometimes with self-reinforcing dynamics and long-lived consequences. Examples investigated in the scientific literature include ice-sheet stability, parts of tropical forests and ocean circulation. The existence of risks does not mean an exact tipping year can be predicted for each system. Risks depend on warming magnitude, duration, regional conditions and interactions. Avoid sensational claims that one unusual season proves an irreversible planetary tipping point. Policy uses the precautionary principle because high-consequence outcomes can justify preventive action despite uncertainty about their precise probability.

Scenario analysis asks what may happen under different future emissions, land-use and socioeconomic pathways. It is not a prediction of a single inevitable future. The IPCC uses scenarios to estimate how warming, precipitation, sea level and impacts change under alternative choices. Model ensembles capture some uncertainty, but structural model limits, natural variability and future human decisions remain. Local projections can be less certain than global warming direction, especially for short-duration extreme rainfall in complex terrain. Vulnerability assessment therefore combines climate scenarios with present-day exposure, social sensitivity and adaptive capacity.

A Telangana planning officer can use a scenario approach to stress-test a water supply: compare demand under population growth, multi-year drought, groundwater depletion and extreme rainfall. Robust measures such as leakage reduction, aquifer recharge, diversified supply and demand management help across several plausible futures. Decisions should be revised as observations improve. In an answer, explain what is known confidently, what is uncertain and which low-regret actions make sense despite uncertainty; do not replace analysis with either fatalism or unsupported technological optimism.

4. Global impacts: heat, water cycle, oceans and cryosphere

Global warming intensifies many heat extremes by shifting the distribution of daily temperatures upward. A modest rise in average temperature can greatly increase the frequency of days beyond a dangerous threshold. Humidity raises physiological risk by slowing evaporative cooling; wet-bulb conditions can be hazardous even when dry-bulb temperature is not record-breaking. Outdoor workers, infants, older people, people with chronic disease and households without cooling or safe water face unequal exposure. Heat can also reduce labour productivity, disrupt electricity demand and damage transport infrastructure.

The water cycle intensifies because a warmer atmosphere can hold more water vapour, but rainfall does not increase evenly everywhere. Heavy rainfall intensity can rise while longer dry spells also occur, depending on circulation, soil and land-use conditions. Flood risk is produced by the intersection of rainfall hazard with drainage capacity, land cover, encroachment, settlement location and emergency preparedness. Urban flooding in Hyderabad cannot be attributed to global warming alone; blocked drains, lost wetlands and impervious surfaces are local contributors. Similarly, a drought is a meteorological and hydrological process influenced by water management and crop choices.

Ocean warming causes thermal expansion, contributes to sea-level rise, shifts marine ecosystems and fuels some marine heatwaves. Melting land ice adds water to the ocean; floating sea-ice melt has a different direct sea-level effect. Ocean uptake of carbon dioxide increases acidity, affecting calcifying organisms and food webs. WMO's 2025 report highlights ocean heat, glaciers and sea ice as key indicators. India’s coastal regions face cyclones, erosion, salinisation and exposure of infrastructure, while landlocked Telangana experiences indirect effects through supply chains, food prices, migration and national fiscal costs.

Mountain glaciers affect downstream seasonal runoff and hazards such as glacial-lake outburst floods. These are important for Indian national planning even though Telangana is not a Himalayan state. Disaster policy should combine early warning, resilient construction, ecosystem protection, social protection and evacuation planning. Avoid claiming every cyclone is caused by climate change; attribution examines changes in probability or intensity, not simplistic one-event causation. An effective mains response groups impacts by hazard, affected sector, vulnerable group and adaptation instrument.

5. Agriculture, food security, livestock and rural livelihoods

Climate impacts on agriculture arise through heat stress, rainfall variability, soil moisture, pests, irrigation availability and extreme events at sensitive crop stages. Higher carbon dioxide can stimulate photosynthesis in some conditions, but gains are constrained by nutrients, water, temperature, pests and crop quality. Heat around flowering can reduce grain setting; unseasonal rainfall can damage harvest and storage; repeated drought can increase indebtedness. Crop yield is not the same as farmer income: prices, input costs, tenancy, access to insurance and market power shape livelihood outcomes. An irrigation project may buffer rainfall risk while increasing groundwater or energy pressure if extraction is unmanaged.

Telangana's rainfed districts illustrate why climate adaptation must be location-specific. Red soils, black soils, tank irrigation, borewell dependence and varied cropping systems create different risk profiles. Cotton, paddy, maize and pulses do not share identical water requirements or heat sensitivities. Advisories from agrometeorological services, sowing-window adjustment, soil-moisture conservation, diversified cropping, drought-tolerant varieties, water budgeting and tank rehabilitation are complementary measures. A shift from water-intensive crops may save water but requires procurement, processing, extension and market support so farmers are not forced to bear transition costs alone.

Livestock experience heat stress through lower milk yields, fertility impacts, increased water demand and disease pressures. Fodder availability becomes a major constraint during drought. Climate-smart animal husbandry includes shade, water, veterinary surveillance, resilient fodder systems and manure management. Methane from enteric fermentation and manure is a mitigation concern, but policies must account for smallholder livelihoods and nutritional needs. Avoid simplistic calls to eliminate livestock without examining social and economic consequences.

Adaptation policy requires coordination among agriculture, irrigation, rural development, finance and weather agencies. Insurance may smooth losses but cannot substitute for risk reduction, and poorly designed claims procedures can exclude tenant farmers. Evaluating a scheme requires tracking uptake by smallholders, actual yield stability, groundwater trends, net income and gender outcomes rather than counting subsidies disbursed. A strong exam answer separates productivity, resilience, mitigation co-benefits and distributive justice.

6. Human health, urban heat islands and inequality

Climate change affects health directly through heat illness, injuries during floods and storms, and indirectly through air quality, water safety, nutrition, vector ecology and mental health. Attribution of a particular illness is difficult because exposure interacts with housing, sanitation, income and access to care. Heat-related illness ranges from dehydration and exhaustion to heatstroke, which is a medical emergency. Workers in construction, street vending, sanitation, agriculture and transport may have high exposure and little control over work schedules. Gender and disability can alter vulnerability through mobility, caregiving and access to services.

The urban heat-island effect is a local phenomenon in which built-up areas retain heat because of dark surfaces, limited vegetation, reduced evaporation and anthropogenic heat release. It interacts with, but is distinct from, global warming. In Hyderabad, tree-canopy loss, paved surfaces, densely built neighbourhoods and unequal housing quality can intensify local heat. Cool roofs, shaded public spaces, urban forestry with appropriate species, drinking-water access, reflective materials, ventilation and emergency response help reduce exposure. Their performance must be measured rather than assumed; a poorly designed reflective intervention may transfer heat to pedestrians.

Heat action plans integrate forecasts, thresholds, public alerts, hospital preparedness, occupational protections and targeted outreach. Telangana's 2025 heatwave planning included cooling and water measures, but implementation quality must be judged using verified district data, hospital preparedness, outreach coverage and excess-mortality monitoring. An early warning that does not reach informal workers is not an effective adaptation system. Schools, care homes and employers need clear response protocols; alerts must use languages and channels accessible to affected groups.

Climate justice recognises that those who contributed least to emissions may face the greatest harms. Within a city, a wealthy household may afford insulation and cooling while a low-income family experiences night-time heat and electricity insecurity. A sound policy package therefore combines climate-resilient infrastructure with labour safeguards, public health, affordable services and grievance mechanisms. In mains answers, move beyond generic 'awareness' to specify responsible institutions, funding, beneficiaries and measurable outcomes.

7. Biodiversity, forests, ecosystem services and carbon sinks

Climate change shifts species distributions, seasonal life cycles, fire regimes and ecological interactions. Some species can migrate or adapt, while fragmented habitats and rapid warming constrain movement. Climate change interacts with habitat loss, invasive species, pollution and overexploitation; these pressures should not be attributed to temperature alone. Ecosystem services include provisioning benefits such as water and food, regulating services such as flood buffering and pollination, cultural benefits and underlying ecological processes. Protecting ecosystems can reduce climate risks while supporting biodiversity and livelihoods, but not every tree plantation is ecologically equivalent to natural forest.

Forest carbon is stored in living biomass, dead matter and soils. Afforestation adds trees to previously non-forested land, while reforestation restores tree cover; neither guarantees permanent carbon storage. Fires, drought, disease, harvesting and land conversion can reverse gains. Carbon-sink accounting needs baseline, additionality, leakage, permanence and independent verification. India’s April 2026 NDC targets a forest-and-tree-cover carbon sink of 3.5–4.0 billion tonnes of CO2 equivalent by 2035 relative to 2005; this is a stated future commitment, not an observed achievement. Conservation of existing biodiverse forests may deliver ecological value beyond carbon tonnes.

Telangana's Kawal and Amrabad landscapes show the overlap between wildlife conservation, forest-dependent livelihoods and climate adaptation. Habitat corridors, invasive-species control, fire preparedness and watershed restoration can enhance resilience. However, conservation policy must respect legal rights and participation of Scheduled Tribes and other forest dwellers under applicable laws. An imposed plantation that displaces access to minor forest produce may shift costs to vulnerable households. Restoration design should distinguish degraded natural forest, grassland, wetland and agricultural commons rather than converting all land to uniform tree cover.

Nature-based solutions require ecological fit and social legitimacy. Urban lakes can reduce flooding and moderate local temperatures, but encroachment, sewage inflow and altered catchments may undermine them. A district-level evaluation should track water quality, biodiversity, flood retention, carbon, maintenance cost and livelihood effects. In a mains answer, present biodiversity protection as a complement to rapid fossil-emission cuts, never as a complete substitute.

8. Mitigation: energy transition, industry and transport

Mitigation addresses the causes of climate change by reducing greenhouse-gas emissions or increasing durable removals. Energy policy is central because coal, oil and gas combustion contributes carbon dioxide. Demand-side efficiency reduces energy needed for a service; supply-side decarbonisation changes the energy source. Electrification can lower emissions when electricity becomes cleaner, but life-cycle emissions depend on generation mix, batteries, materials and use patterns. Solar and wind power require planning for variability, transmission, storage, flexible demand and grid stability. Installed capacity should not be confused with annual electricity generation.

Industrial mitigation involves efficiency, waste-heat recovery, cleaner process fuels, material efficiency, recycling, low-carbon electricity and innovation in hard-to-abate processes such as cement and steel. Carbon capture may be relevant in some sectors, but its cost, energy demand, storage permanence and actual capture rates require scrutiny. Circular economy measures can reduce embodied emissions through repair, reuse, remanufacturing and lower material throughput. Avoid treating a firm's renewable-energy certificate or a 'net-zero' label as proof of zero physical emissions without examining boundaries and offsets.

Transport choices involve public transport, compact urban design, walking and cycling safety, electrified fleets, fuel efficiency and freight logistics. Electric vehicles reduce tailpipe emissions but do not eliminate congestion, road injuries, tyre particles or lifecycle impacts. Hyderabad's mobility choices demonstrate the importance of combining cleaner vehicles with reliable mass transit and last-mile access. Land-use planning that reduces travel distance can outperform a technology-only strategy. Household energy and buildings require insulation, efficient appliances, passive cooling and accessible finance; solutions should not transfer cost burdens to low-income residents.

Mitigation policy mixes regulation, standards, public procurement, pricing, infrastructure, research and information. A carbon price can incentivise cleaner choices but may require rebates or targeted protection for poor households. A subsidy can accelerate adoption but may also create fiscal leakage or lock-in if poorly targeted. Evaluate measures using emissions reduced, cost per tonne, reliability, jobs, air quality, equity and implementation capacity. A just transition includes worker retraining and support for coal-dependent regions rather than assuming new jobs automatically appear where old jobs disappear.

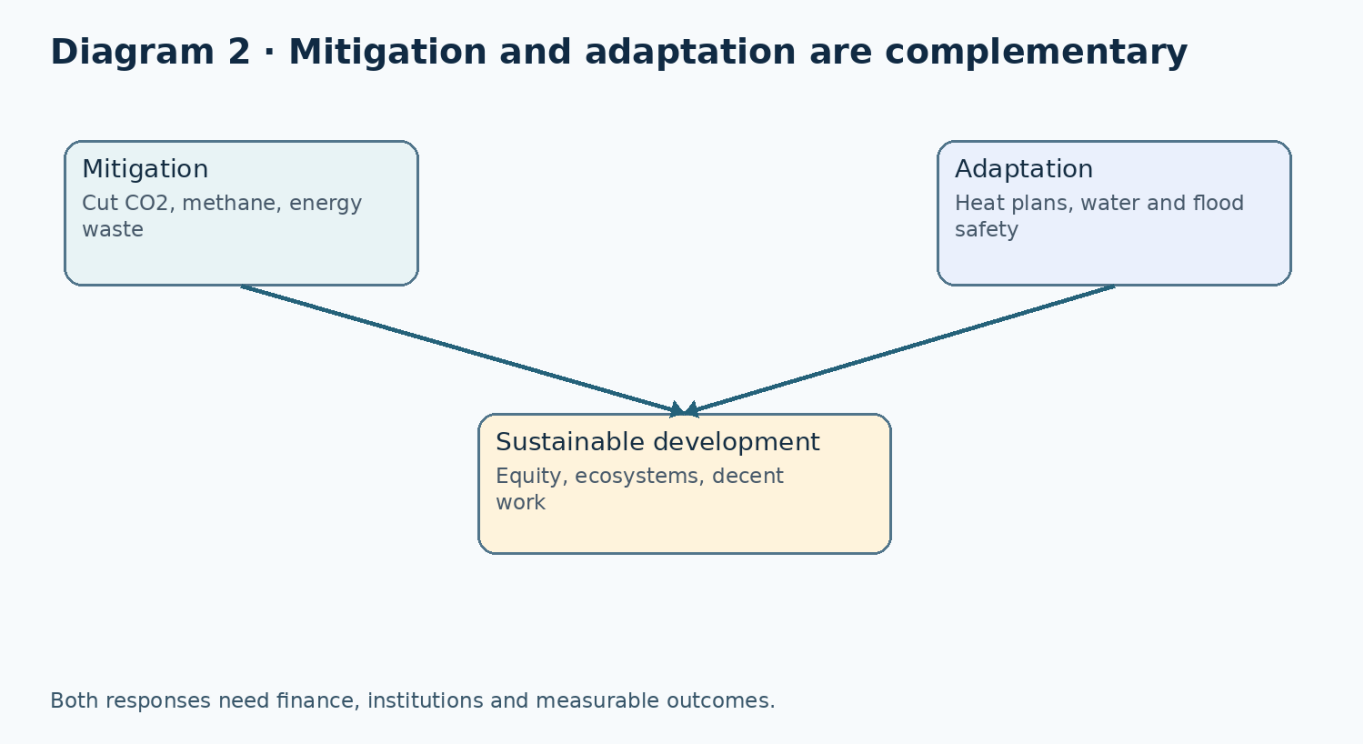
9. Adaptation, resilience, disaster risk and loss and damage

Adaptation is adjustment to actual or expected climate effects; mitigation reduces their underlying drivers. The same action can serve both purposes, but they must not be conflated. A restored wetland can store carbon and reduce floods, while a heat alert reduces exposure without materially reducing global emissions. Resilience refers to capacity to anticipate, absorb, recover and transform after shocks. Disaster risk combines hazard, exposure, vulnerability and capacity. A higher flood level is a hazard; housing in a floodplain creates exposure; weak construction and lack of savings increase vulnerability.

Adaptation options include climate-resilient water infrastructure, drought planning, resilient crops, floodplain zoning, early warning, disease surveillance, cool roofs and social protection. Their effectiveness varies by context and may decline as warming increases. Maladaptation occurs when an intervention reduces immediate risk for one group but increases long-term vulnerability, emissions or inequity. For example, unlimited borewell pumping during drought may protect this year's crop while depleting aquifers; embankments may protect one locality while increasing downstream flooding. Risk-informed appraisal must account for such spillovers.

Loss and damage refers to adverse climate impacts that occur despite, or beyond the limits of, mitigation and adaptation. Economic losses include damaged assets and lost production; non-economic losses can involve cultural heritage, health, displacement and ecosystem values. International negotiations address finance and arrangements for vulnerable countries, but these mechanisms are not a substitute for domestic risk reduction. At district level, accurate exposure maps, early warning, evacuation drills, resilient schools and transparent relief systems are immediate governance priorities.

Telangana's drought-prone agricultural areas and flood-prone urban catchments need different adaptation portfolios. Mandal-level vulnerability mapping should include meteorological observations, water sources, occupational exposure, health services, settlement patterns and social vulnerability. Plans need clear triggers and budgets: who issues alerts, who opens shelters, how farmers receive advisories, and how results are audited. In a mains answer, recommend actions at household, community, local-government and state levels, and identify metrics such as heat morbidity, groundwater balance and flood-loss trends.



Original diagram 2: 02 policy response.png.

10. Global climate governance: UNFCCC, Kyoto and Paris

The United Nations Framework Convention on Climate Change (UNFCCC), adopted in 1992, establishes a framework for international cooperation to stabilise greenhouse-gas concentrations while recognising development needs. Its principle of common but differentiated responsibilities and respective capabilities (CBDR-RC) acknowledges unequal historical contributions and capacities. The Kyoto Protocol (1997) imposed quantified commitments on certain developed-country parties under its architecture; its scope and participation limits contributed to calls for a more universal arrangement. The Paris Agreement, adopted in 2015 and entered into force in 2016, brought all parties into a system of nationally determined contributions (NDCs).

Article 2 of the Paris Agreement aims to hold global average temperature increase well below 2°C above pre-industrial levels while pursuing efforts toward 1.5°C, strengthen adaptation and align financial flows with low-emission, climate-resilient development. Countries communicate and update NDCs, with progression expected over time. The enhanced transparency framework supports reporting and review; the global stocktake assesses collective progress, not the guilt or compliance of a single state. The first global stocktake concluded at COP28 in 2023 and called for accelerated action, including transitioning away from fossil fuels in energy systems. Treaty goals, COP decisions and national pledges have different legal and operational status.

Climate finance includes grants, concessional lending, private capital and other instruments, but debates concern additionality, access, debt burdens and the balance between mitigation and adaptation. Technology transfer is more than shipping equipment: intellectual property, local skills, maintenance, finance and institutional capacity affect use. Equity disputes include historical emissions, per-capita emissions, remaining carbon budget, loss and damage and development space. Developed countries emphasise broad participation; developing countries emphasise historical responsibility and predictable support. An exam answer should present both without implying that equity negates the physics of warming.

Other agreements complement climate governance. The Montreal Protocol addresses ozone-depleting substances, and the Kigali Amendment phases down HFCs with climate benefits; ozone depletion and global warming are distinct scientific problems. The Convention on Biological Diversity, UN Convention to Combat Desertification and Sendai Framework intersect with resilience. Avoid listing international treaties as interchangeable. Evaluate each by its objective, commitments, implementation mechanism, finance and observed outcomes.

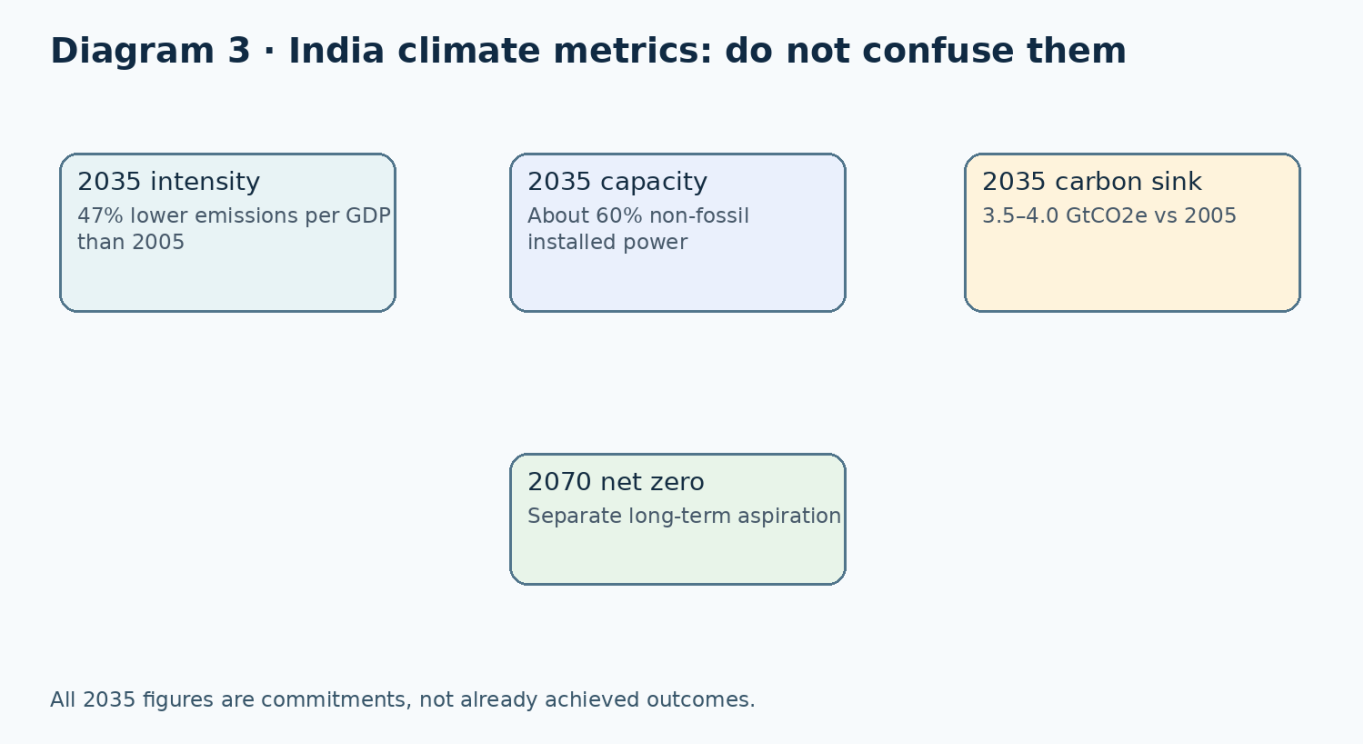
11. India: NAPCC, NDCs, net zero and policy architecture

India's National Action Plan on Climate Change (NAPCC), launched in 2008, originally comprised eight national missions covering solar energy, enhanced energy efficiency, sustainable habitat, water, sustaining Himalayan ecosystems, Green India, sustainable agriculture and strategic knowledge. Later government descriptions may count a health mission as an additional mission; always identify the source and year when stating the total. State Action Plans on Climate Change adapt national objectives to regional risks. Ministries, state departments, electricity regulators, municipalities, agricultural institutions and disaster authorities implement different parts of the response. India's long-term net-zero goal is 2070, a future objective rather than current zero emissions.

India updated its NDC in 2022 and submitted an NDC for 2031–2035 to the UNFCCC on 24 April 2026. The official April 2026 document commits to reducing the emissions intensity of GDP by 47% by 2035 relative to 2005, reaching about 60% cumulative installed electric-power capacity from non-fossil sources by 2035 with support, and creating a 3.5–4.0 billion tonne CO2-equivalent forest-and-tree-cover carbon sink by 2035 relative to 2005. It also emphasises adaptation, Mission LiFE, climate finance and technology cooperation. The document explicitly states that it does not impose sector-specific emissions-reduction obligations, including for agriculture. These are NDC commitments, not proof that 2035 outcomes have occurred.

The same NDC reports that non-fossil sources represented 52.57% of installed electric-power capacity on 28 February 2026. This is an observed capacity statistic for that date, not a claim about the share of electricity generated. It also states a 36% fall in GDP emissions intensity between 2005 and 2020. A falling intensity does not guarantee falling absolute emissions. A sound answer distinguishes policy instruments such as energy-efficiency programmes, renewable procurement, building standards and clean transport from national target metrics. India's carbon market architecture and regulatory details should be verified against current notifications before claiming any particular sector is already covered.

India's climate strategy faces legitimate trade-offs: development and affordable energy, reliability and decarbonisation, industrial competitiveness and air quality, public finance and climate adaptation. Solar manufacturing, battery supply chains and transmission networks have material and land impacts that require governance. A practical transition uses least-cost planning, energy access, worker support, transparent environmental assessment and distributional safeguards. Mains answers should explicitly separate 2030 commitments, 2035 commitments, the 2047 development vision and the 2070 net-zero aspiration.



Original diagram 3: 03 commitments metrics.png.

12. Telangana climate governance, water, agriculture and cities

Telangana's Environment Protection Training and Research Institute (EPTRI) serves as a climate-change knowledge and coordination institution. Its official climate page states that GO Ms No 06 dated 17 March 2016 appointed EPTRI as

the nodal agency for the Clean Development Mechanism and Centre for Climate Change. It reports that the Telangana State Action Plan on Climate Change was prepared in 2015 and is under revision. The Telangana State Climate Change Centre supports risk assessment, data repositories, sectoral adaptation and capacity building. A plan's existence is not evidence of successful implementation; implementation must be assessed through budgets, project outcomes and vulnerability indicators.

Heat exposure, monsoon variability, tank and reservoir management, groundwater stress, forest fire risk and urban drainage are important Telangana pathways. Hyderabad's urban heat island combines with broader warming but also depends on land use, tree cover, reflective surfaces and building density. Urban flood resilience requires restoring drainage channels and lakes, enforcing land-use rules and maintaining infrastructure. Rural water resilience depends on aquifer conditions, irrigation demand, watershed restoration and crop choice. Because different districts face different hazards, a state-wide average can conceal severe local vulnerability.

EPTRI describes the RAH-ACT initiative for climate-resilient agricultural households in the erstwhile Mahabubnagar district, involving agriculture institutions and ICRISAT, and a landscape-restoration initiative across multiple districts. These are reported programme descriptions, not independent evidence of measured income or ecological outcomes. A district administration should examine baseline conditions, beneficiary selection, maintenance and follow-up measurements. Adaptation should prioritise marginal farmers, tenant cultivators, outdoor workers, informal settlements and forest-dependent households, with accessible grievance channels and gender-sensitive design.

A Telangana climate budget could tag expenditure by mitigation, adaptation and general development, but tagging alone cannot demonstrate additional climate impact. Indicators might include heat-related hospital visits, percentage of vulnerable wards with cooling plans, aquifer recharge versus extraction, flood-disruption hours, tree survival and crop-income volatility. Use time-series evidence and independent evaluation before claiming programme success. In an exam answer, anchor state action in EPTRI, the State Action Plan, disaster management, urban local bodies and agricultural extension; then identify coordination failures and specific remedies.

13. Sustainable development: origins, pillars and ethics

The Brundtland Commission's 1987 report popularised sustainable development as meeting present needs without compromising the ability of future generations to meet their own needs. The concept is not simply 'growth with a few trees'. It integrates economic opportunity, social justice and ecological integrity while recognising biophysical limits. Intergenerational equity concerns future people; intragenerational equity concerns unequal opportunities and burdens among people alive today. The precautionary principle supports preventive action where serious or irreversible harm is plausible despite scientific uncertainty; the polluter-pays principle places costs of pollution control and damage on responsible actors.

Three dimensions are analytically distinct but connected. Economic sustainability involves productive capacity, fiscal stability and livelihoods. Social sustainability includes health, education, equality, rights and participation. Environmental sustainability requires protection of ecosystems, natural capital and pollution-absorbing capacity. A project that raises GDP while displacing vulnerable people and exhausting groundwater may fail a sustainable-development test. Conversely, environmental regulation that ignores employment and affordability can produce social resistance and inequity. The policy task is to design institutions that address these tensions transparently rather than pretending all objectives automatically align.

Weak sustainability assumes many forms of natural capital can be replaced by human-made capital; strong sustainability argues that critical ecological functions have limited substitutes. Wetlands, stable climate, aquifers and biodiversity can exhibit thresholds or irreversible loss. This distinction matters when appraising a highway, mine, irrigation dam or industrial estate. Cost-benefit analysis can illuminate trade-offs but may undervalue non-market services and distributive harms. Environmental impact assessment, strategic assessment, public consultation and long-term monitoring are therefore governance complements to financial appraisal.

Sustainable development requires indicators beyond GDP: health, learning, poverty, emissions, resource productivity, water security, ecological condition and resilience. Metrics should be disaggregated by gender, location, income and social group. In Telangana, a sustainable urban plan would jointly assess transport access, affordable housing, heat exposure, lake health and municipal finance. An exam answer should connect the definition to principles, institutional tools, trade-offs and measurable outcomes instead of ending with a slogan.

14. SDGs, indicators and policy coherence

The United Nations 2030 Agenda, adopted in September 2015, sets out 17 Sustainable Development Goals and 169 targets. These goals are integrated: SDG 13 addresses climate action; SDG 7 energy; SDG 6 water; SDG 11 sustainable cities; SDG 12 responsible consumption; SDG 15 terrestrial ecosystems; SDG 2 food systems; and SDG 3 health. Goal 17 concerns finance, technology, partnerships and data. Climate action can advance other goals through cleaner air and energy access, but trade-offs arise when large projects displace communities or strain land and water. A goal is a normative commitment; a target specifies an outcome; an indicator measures progress. These are not interchangeable.

The Agenda's 'leave no one behind' principle means aggregate improvement is insufficient if the most vulnerable groups remain excluded. A city can increase average piped-water access while informal settlements receive unreliable service. Likewise, a state can expand renewable power while workers in fossil-dependent supply chains lose livelihoods. Policy coherence examines whether agriculture subsidies, water tariffs, urban zoning, industrial approvals and energy incentives reinforce or undermine one another. Fragmented departmental targets often cause perverse outcomes: subsidised pumping may increase crop production today but weaken future water security.

Indicators need careful interpretation. A high share of installed renewable capacity says little about generation without utilisation data. A rising tree-planting count says little about survival, species suitability or natural forest quality. Poverty reduction may improve resilience, but it cannot replace hazard-specific preparedness. Climate indicators should use consistent geographic boundaries, baselines and reporting dates. State SDG dashboards are useful for tracking but cannot automatically prove causal effects of a policy. Independent evaluation asks what would have happened without the intervention.

For Telangana, integrated planning could link watershed restoration, climate-resilient crops, women's self-help groups, urban transport and primary health systems. Finance departments can demand that major proposals specify expected SDG benefits, possible adverse effects and monitoring responsibilities. Gram sabhas and urban ward committees can surface local impacts hidden by statewide averages. In a mains answer, choose a few linked goals and explain mechanisms rather than reciting all seventeen without analysis.

15. Climate finance, carbon markets, equity and just transition

Climate finance funds mitigation, adaptation and responses to losses. Public finance can build early-warning systems, public transport and resilient water infrastructure; private capital may support clean generation or industrial efficiency when revenue models are credible. Grants are often important for vulnerable communities and public goods that cannot repay loans. Concessional loans, guarantees and blended finance can mobilise investment but may create debt or contingent fiscal liabilities. A financing announcement is not the same as money disbursed, and a climate-tagged project is not automatically additional to pre-existing spending.

Carbon pricing can take the form of a tax or emissions trading system. A carbon tax fixes a price while emissions respond; a cap-and-trade system fixes an aggregate quantity while allowance prices vary, subject to design. Both require credible monitoring, reporting and verification. Offset credits claim reductions or removals outside a buyer's own operations; quality depends on additionality, permanence, leakage, baseline choice and double-counting controls. Forest offsets are particularly exposed to fire and reversal risk. Carbon markets may lower mitigation costs but cannot replace direct regulation of local pollution or robust public accountability.

Climate justice has international and domestic dimensions. Countries differ in historical cumulative emissions, present per-capita emissions and capacities; within countries, income groups and regions differ. The principle of CBDR-RC guides international negotiations but does not excuse avoidable domestic pollution or poor adaptation. A just transition protects workers and communities during economic restructuring through social dialogue, retraining, regional investment, wage support and access to new jobs. A coal-dependent district may need different policies from a fast-growing technology hub.

Practical fiscal choices involve cost-effectiveness, affordability and resilience. For a Telangana municipality, flood-drain maintenance and shade for public spaces may offer high immediate benefits even if they do not generate marketable carbon credits. For a state utility, grid investment may unlock renewable energy while maintaining reliability. Public procurement can create demand for efficient equipment, but contracts need measurable performance. Mains answers should distinguish who pays, who benefits, who bears risk and how outcomes will be verified.

16. Integrated policy choices, controversies and answer strategy

Climate policy debates often present false binaries: development versus environment, mitigation versus adaptation, or technology versus behaviour. Real policy must integrate them while acknowledging trade-offs. A dam may improve drought resilience yet alter downstream ecology; solar parks may cut emissions but affect grazing land; air-conditioning can save lives during heatwaves while increasing peak electricity demand. A high-quality evaluation asks about alternative designs, beneficiaries, lifecycle effects, local rights, fiscal costs and reversibility. Public participation improves legitimacy but does not remove the need for technical analysis.

Climate scepticism sometimes exploits uncertainty about a particular local forecast to deny well-established global warming. The opposite error is to attribute every disaster entirely to climate change without examining exposure and governance. Scientific reasoning separates observed trend, physical mechanism, event attribution and future scenario. Policy claims require the same discipline: a government target is not an achievement; a new law is not evidence of enforcement; and a pilot is not proof of statewide effectiveness. Date and denominator are essential whenever numerical facts appear.

An effective district climate action plan begins with a baseline: hazard maps, vulnerable populations, sectoral emissions and institutional capacity. It prioritises measures using avoided harm, cost, equity and feasibility, assigns responsibilities, sets financing and maintenance plans, and establishes monitoring. For Telangana, priorities could include heat-health protection, lake and drainage restoration, efficient irrigation, resilient public services, distributed clean energy and reliable public transport. These actions should be adapted to local ecological conditions and implemented with meaningful community participation.

For a 200–250-word mains answer, use four or five short subheadings: context, mechanisms, Telangana application, governance measures and conclusion. Include one verified dated fact when useful, but avoid stuffing the answer with unrelated statistics. Distinguish mitigation, adaptation and sustainable development, connect science to institutions and end with an implementable recommendation. When an official PYQ is not available with a source paper and question number, label the question as original practice. This chapter is a published learning draft, not an independently approved academic edition.

Telangana and policy case-study applications

Case 1: Hyderabad heat and built environment

A city heat-risk intervention should distinguish background warming from the urban heat-island effect. Mapping air temperature alone misses household exposure because building materials, ventilation, roof insulation, electricity reliability, outdoor work and nighttime heat affect actual risk. Satellite land-surface temperature helps locate hotspots but is not the same as near-surface air temperature experienced by people. A rigorous pilot could compare shaded bus stops, reflective roofs, water points and work-rest guidance in similar wards, tracking usage, indoor temperature and heat illness. Urban forestry requires water budgeting and tree-survival monitoring, not simply planting totals. A fair policy protects street vendors, construction workers and elderly residents rather than focusing only on commercial districts.

Case 2: Rainfed farming and groundwater stress

A Telangana mandal may face delayed monsoon onset, a few intense rainfall events and a long dry spell in the same season. The policy response must integrate sowing advisories, soil moisture, tanks, borewell management, crop insurance and markets. More irrigation alone can be maladaptive when groundwater extraction exceeds recharge. Farmers cannot shift crops safely without procurement, storage, extension and income-risk support. Compare outcome indicators across seasons: yield stability, net farm income, borewell failure, aquifer levels and household debt. Rainfall totals are an inadequate sole measure because distribution and timing determine crop stress. Participation by tenant cultivators and women farmers is necessary for accurate diagnosis.

Case 3: Urban flood resilience and catchment governance

A high-intensity rain event becomes a disaster when drainage channels are blocked, lakes lose storage, settlements occupy low-lying areas and emergency communication fails. Attribution studies may assess whether warming increased the probability or intensity of rainfall, but local decisions determine much of the damage. An integrated urban catchment

plan inventories inlets, culverts, wetlands, impervious surfaces and vulnerable households; coordinates upstream and downstream jurisdictions; and tests evacuation routes. Engineering solutions should be compared with nature-based retention, land-use enforcement and maintenance. Success is measured through reduced flood depth, shorter disruption, fewer injuries and fair compensation, not the length of drains constructed.

Case 4: Forest restoration and community rights

A forest-carbon project in Telangana must establish whether land is degraded forest, natural grassland, wetland or community-use landscape. Planting fast-growing monocultures can generate a short-term biomass statistic while damaging biodiversity, water balance or customary livelihoods. Restoration should favour ecological suitability, habitat connectivity and species diversity, with rights recognition and local participation. Carbon claims need independent measurement, leakage checks, permanence assessment and transparent benefit-sharing. Community forest management can support fire prevention and livelihood resilience when incentives are credible. Evaluation should compare baseline biodiversity, canopy quality, soil moisture, fire occurrence and household benefits rather than reporting seedlings alone.

Case 5: Just transition and energy planning

A credible energy transition in a coal-dependent economy must maintain affordable electricity while lowering emissions. Installed renewable capacity alone is not enough: the grid requires transmission, storage, flexible demand, forecasting and reliability planning. Workers and suppliers connected to existing fossil-fuel infrastructure may need retraining, social security and alternative local investment. A new solar project can create jobs but not necessarily in the same district or with the same skill requirements. Land acquisition and recycling of panels and batteries create environmental obligations. Public evaluation should report actual electricity generation, emissions intensity, outage frequency, tariff impact, employment quality and local environmental safeguards.

Worked analytical applications and examination traps

Application 1: Worked analytical problem: emissions intensity versus total emissions

Suppose an economy emits 100 units of greenhouse gases and produces GDP of 100 monetary units in a baseline year, giving intensity 1 emission unit per GDP unit. Ten years later, emissions rise to 110 while GDP rises to 150. Intensity is 110/150 = 0.733, a reduction of approximately 26.7%, even though absolute emissions increased by 10%. Therefore, a 47% intensity-reduction target cannot be interpreted as a 47% absolute-emissions reduction. The distinction matters when comparing economic growth strategies and climate outcomes. An assessment must also specify whether emissions include land use, which gases are counted, how GDP is measured, and whether data are comparable across years. In a mains answer, present the formula, a numerical illustration and a clear policy implication: intensity improvements should be accompanied by credible analysis of absolute emissions and long-term net-zero pathways.

Application 2: Worked analytical problem: heat risk and social vulnerability

Consider two Hyderabad wards with identical afternoon air temperature. Ward A has shaded streets, insulated roofs, reliable electricity, clinics and public water points. Ward B has tin-roof dwellings, high-density informal employment, little shade and poor access to care. Meteorological hazard may be similar, but exposure and vulnerability differ sharply. A heat-action plan that sends the same text message to both wards may not produce equitable protection. A better design identifies outdoor occupations, night-time indoor temperature, language access, disability and emergency transport. Prioritisation should be transparent and privacy-preserving. Measure implementation through coverage of high-risk households, water availability, cooling-centre usage and heat-related hospital admissions. This example demonstrates why a hazard forecast is necessary but insufficient: climate resilience depends on social and institutional capacity.

Application 3: Worked analytical problem: carbon sink and ecological integrity

A district proposes a plantation to claim carbon removals. Before accepting the claim, an evaluator asks whether the land would naturally regenerate without the project (additionality), whether fire or land-use change could reverse storage (permanence), whether grazing or wood extraction is displaced elsewhere (leakage), and whether the carbon is counted twice. The evaluator also asks whether native habitat, grassland biodiversity, water balance and community rights are protected. The carbon benefit should be measured against a credible baseline over time rather than inferred from sapling purchase invoices. A project can improve carbon storage yet still harm ecosystem services if it replaces natural grassland with a water-intensive monoculture. Good sustainable-development appraisal treats biodiversity and rights as substantive outcomes, not optional co-benefits.

Application 4: Worked analytical problem: adaptation cost-effectiveness

Suppose a municipality can finance either a high-capital flood barrier or a portfolio of drain maintenance, wetland restoration, early warning and targeted relocation. The first may deliver visible protection for one zone but shift runoff downstream and require expensive upkeep. The second may distribute benefits across catchments but demand interdepartmental coordination and land-use enforcement. A robust appraisal compares expected annual damage avoided, distribution of benefits, maintenance liabilities, ecological consequences and performance under multiple rainfall scenarios. It should also consider non-monetised losses, such as interrupted schooling and disease risk. The least initial cost is not necessarily the lowest lifecycle cost. Monitoring must test whether promised risk reductions actually occur and whether vulnerable residents have access to warnings and compensation.

Application 5: Worked analytical problem: SDG policy coherence

A programme subsidises electricity for irrigation to improve short-run farm income, supporting food production and poverty reduction. Without groundwater controls, however, it may accelerate aquifer depletion and undermine SDG 6 water security, SDG 13 adaptation and future agricultural productivity. An integrated reform could combine targeted income support, efficient irrigation, water accounting, crop diversification and market access. It should not punish farmers without alternatives. Evaluation compares income distribution, energy use, groundwater trends, food security and fiscal cost over multiple seasons. This illustrates why the 17 SDGs are described as integrated: progress on one indicator can conceal deterioration elsewhere. A high-quality policy note identifies complementary instruments and trade-offs rather than assuming all SDG targets move together.

Consolidated chapter summary

Climate change is an Earth-system problem with an unambiguous human driver but heterogeneous regional effects. Global warming alters energy balance, hydrology, cryosphere, ecosystems, health and livelihoods; vulnerability determines who suffers most. Mitigation and adaptation are complementary and must be embedded in sustainable development rather than framed as substitutes. International governance uses the UNFCCC, Kyoto and Paris frameworks; India has differentiated 2035 intensity, capacity and sink commitments alongside its 2070 net-zero goal. Telangana must translate this architecture into heat protection, resilient farming, urban flood management, water security, ecosystem restoration and just institutions. Good policy separates observed data from projections and targets, and measures actual social and ecological outcomes.

Mains answer blueprints

Blueprint 1: Explain the scientific basis of global warming and why one hot year does not define the Paris temperature threshold.

Suggested sequence: Context or definition → physical/economic mechanisms → evidence or Telangana application → governance responses → balanced conclusion. **Avoid:** mixing targets and outcomes, claiming every event is climate-caused, or giving an unstructured single paragraph.

Blueprint 2: Assess the climate-change risks for Telangana agriculture and recommend a practical adaptation strategy.

Suggested sequence: Context or definition → physical/economic mechanisms → evidence or Telangana application → governance responses → balanced conclusion. **Avoid:** mixing targets and outcomes, claiming every event is climate-caused, or giving an unstructured single paragraph.

Blueprint 3: Compare mitigation and adaptation with examples from Hyderabad and Telangana.

Suggested sequence: Context or definition → physical/economic mechanisms → evidence or Telangana application → governance responses → balanced conclusion. **Avoid:** mixing targets and outcomes, claiming every event is climate-caused, or giving an unstructured single paragraph.

Blueprint 4: Examine India’s 2031–2035 NDC and distinguish commitments from observed achievements.

Suggested sequence: Context or definition → physical/economic mechanisms → evidence or Telangana application → governance responses → balanced conclusion. **Avoid:** mixing targets and outcomes, claiming every event is climate-caused, or giving an unstructured single paragraph.

Blueprint 5: Discuss sustainable development and the policy tensions between growth, equity and ecology.

Suggested sequence: Context or definition → physical/economic mechanisms → evidence or Telangana application → governance responses → balanced conclusion. **Avoid:** mixing targets and outcomes, claiming every event is climate-caused, or giving an unstructured single paragraph.

Blueprint 6: Evaluate international climate governance and the importance of climate justice for developing countries.

Suggested sequence: Context or definition → physical/economic mechanisms → evidence or Telangana application → governance responses → balanced conclusion. **Avoid:** mixing targets and outcomes, claiming every event is climate-caused, or giving an unstructured single paragraph.

Thirty original explained MCQs (not official PYQs)

MCQ 1. Which statement correctly distinguishes climate from weather?

A. Climate is a single day of atmospheric conditions

B. Climate describes long-term distributions of weather

C. Weather is always regional while climate is always global

D. Climate excludes variability

Answer: B. Climate describes statistical characteristics and variability over extended periods; a single hot or cold day is weather. Climate analysis uses defined baselines and spatial scales.

MCQ 2. Which gas is best described as a feedback rather than the principal anthropogenic forcing?

A. Carbon dioxide

B. Methane

C. Water vapour

D. Nitrous oxide

Answer: C. Water vapour increases as air warms and amplifies warming; anthropogenic emissions of long-lived greenhouse gases drive the initial enhanced greenhouse effect.

MCQ 3. What does a positive radiative forcing generally indicate?

A. A cooling tendency

B. An added warming influence on the energy balance

C. No change in ocean heat

D. A fall in greenhouse gases

Answer: B. Positive forcing increases net energy retained by the climate system, although the surface response is delayed by ocean heat uptake and other feedbacks.

MCQ 4. What is the most defensible interpretation of a single year above 1.5°C?

A. The Paris Agreement has legally ceased

B. Long-term warming is automatically above 1.5°C

C. It is a warning signal but not alone the long-term Paris metric

D. Global warming has ended

Answer: C. Paris temperature goals refer to sustained long-term change. Annual variability can produce a single-year exceedance before a long-term threshold is assessed.

MCQ 5. Which observation most directly captures accumulated planetary heat?

A. Daily minimum temperature in one city

B. Ocean heat content

C. One winter snowfall

D. Single cyclone landfall

Answer: B. The ocean stores most excess energy and ocean heat content integrates warming across years, making it a powerful long-term climate indicator.

MCQ 6. Which mechanism is an example of positive climate feedback?

A. Melting ice exposes darker water that absorbs more sunlight

B. Cooling reduces snow cover

C. Volcanic aerosols increase solar input

D. Urban shade increases surface absorption

Answer: A. The ice–albedo feedback reinforces initial warming because dark water absorbs more solar energy than reflective ice.

MCQ 7. Which claim about climate tipping points is scientifically careful?

A. Every tipping threshold is known to a calendar year

B. They are possible state shifts with uncertain thresholds

C. Any flood proves a tipping point

D. All tipping elements have already shifted

Answer: B. Some Earth-system components may change nonlinearly; thresholds and timings are uncertain, so policy combines precaution with ongoing observation.

MCQ 8. Why can warming increase both drought and heavy rainfall risks?

A. Rainfall must rise equally everywhere

B. Circulation and moisture availability change unevenly

C. Evaporation stops in warm climates

D. All regions become uniformly wet

Answer: B. A warmer atmosphere can carry more moisture, but circulation, land conditions and seasonality determine where intense rain or prolonged dryness occurs.

MCQ 9. Which is NOT a sound inference from a rise in renewable installed capacity?

A. The power system may need more flexibility

B. Grid integration matters

C. Renewable electricity generation share must equal capacity share

D. Storage can aid balancing

Answer: C. Capacity measures maximum rated output, not energy produced. Capacity factors, curtailment and dispatch determine actual generation.

MCQ 10. An emissions-intensity target is measured as

A. Total emissions without denominator

B. Emissions divided by GDP

C. Installed solar capacity divided by population

D. Forest area divided by GDP

Answer: B. Emissions intensity expresses emissions per unit economic output; total emissions may still grow when GDP grows faster than intensity falls.

MCQ 11. Which choice is an adaptation measure?

A. Reducing methane leakage from gas systems

B. Issuing heat-health alerts and opening cooling centres

C. Replacing coal with solar solely to cut CO2

D. Capturing cement process emissions

Answer: B. Heat-health alerts lower exposure to an existing climate hazard; the other choices principally target the emissions driving future warming.

MCQ 12. Which is a classic maladaptation risk?

A. Repairing urban drains

B. Unlimited drought-time groundwater extraction without recharge planning

C. Monitoring hospital heat cases

D. Protecting natural wetlands

Answer: B. Unregulated extraction may buffer one season but deplete aquifers and increase future vulnerability, especially for households with shallow wells.

MCQ 13. Which Indian state-level institution is identified as a Telangana climate nodal agency?

A. Reserve Bank of India

B. EPTRI

C. Election Commission of India

D. NHAI

Answer: B. Telangana government designated EPTRI as a climate-change nodal institution through a 2016 government order; implementation spans multiple departments.

MCQ 14. What is a heat island?

A. Global greenhouse warming only

B. Local urban warming linked to built form and land cover

C. A tropical cyclone over a city

D. A seasonal ocean-current reversal

Answer: B. Urban materials, reduced vegetation and human heat release can make built-up areas hotter than surrounding areas; this interacts with global warming.

MCQ 15. Which is a suitable indicator for heat adaptation?

A. Number of slogans printed

B. Measured heat illness and vulnerable-population coverage

C. Installed solar capacity alone

D. Number of saplings purchased

Answer: B. Outcome-oriented monitoring should assess exposure reduction, hospital cases and whether alerts and cooling measures reach high-risk groups.

MCQ 16. Which statement about forest-carbon projects is correct?

A. Seedling count proves permanent removals

B. Additionality and permanence matter for credible carbon claims

C. All monocultures equal natural forests

D. Community rights are irrelevant

Answer: B. Carbon claims require baseline comparisons, survival and reversal risk, leakage checks and independent verification, alongside ecological and social safeguards.

MCQ 17. Which statement best describes the UNFCCC principle CBDR-RC?

A. All countries have identical historical responsibility

B. Responsibilities and capacities differ across countries

C. Only developing countries must act

D. Climate policy excludes equity

Answer: B. CBDR-RC recognises unequal contributions and capabilities while supporting collective climate action; it informs debates on finance and fair burden-sharing.

MCQ 18. What is the Paris Agreement temperature goal?

A. Exactly 3°C

B. Well below 2°C while pursuing 1.5°C

C. No warming relative to 2020

D. A 1.5°C annual cap in every country

Answer: B. Article 2 sets a global long-term temperature objective, not an annual country-specific temperature limit.

MCQ 19. Which statement about NDCs is correct?

A. They are the same as IPCC predictions

B. They are nationally communicated climate commitments

C. They measure only forest cover

D. They apply only to the Kyoto Protocol

Answer: B. Nationally determined contributions describe countries' climate plans under the Paris architecture and are subject to communication and transparency processes.

MCQ 20. India's 2035 GDP emissions-intensity target relative to 2005 is

A. 15% reduction

B. 47% reduction

C. 100% absolute reduction

D. 60% reduction

Answer: B. The April 2026 NDC specifies a 47% emissions-intensity reduction by 2035 from the 2005 baseline; it is not a 47% cut in absolute emissions.

MCQ 21. India's 2035 non-fossil target refers to

A. Total final energy consumption

B. Installed electric-power capacity

C. All vehicle sales

D. Forest area

Answer: B. The official NDC targets about 60% of cumulative installed electric-power capacity from non-fossil resources by 2035, subject to stated support conditions.

MCQ 22. What does India's 2070 commitment describe?

A. A completed achievement in 2026

B. A long-term net-zero goal

C. The date the Paris Agreement started

D. A target for forest area alone

Answer: B. Net zero by 2070 is a future economy-wide aspiration and should not be confused with 2035 NDC intensity or capacity commitments.

MCQ 23. Which goal most directly addresses climate action?

A. SDG 13

B. SDG 4

C. SDG 16

D. SDG 9

Answer: A. SDG 13 is Climate Action; related goals address energy, water, cities, consumption, ecosystems and partnerships.

MCQ 24. How many SDGs and targets are in the 2030 Agenda?

A. 8 and 21

B. 17 and 169

C. 20 and 100

D. 12 and 150

Answer: B. The UN 2030 Agenda adopted in 2015 establishes 17 integrated Sustainable Development Goals and 169 targets.

MCQ 25. Which principle addresses obligations to future generations?

A. Intergenerational equity

B. Opportunity cost alone

C. Strictly short-run efficiency

D. Non-intervention

Answer: A. Intergenerational equity requires that present decisions not unfairly undermine the opportunities and environmental conditions of future people.

MCQ 26. Which is a key concern with carbon offsets?

A. They always cut fossil emissions at source

B. Additionality, leakage and permanence

C. They have no baseline requirements

D. They are equivalent to all forms of adaptation

Answer: B. An offset credit needs a credible counterfactual, no double counting and durable benefit; poor-quality credits can exaggerate real climate progress.

MCQ 27. What is a just transition?

A. Ending all employment in fossil sectors immediately

B. Supporting workers and communities during structural change

C. Replacing every bus with a private car

D. Counting all green investment as adaptation

Answer: B. A just transition combines decarbonisation with social dialogue, worker protection, retraining and regional economic alternatives.

MCQ 28. Which approach best evaluates a flood-resilience project?

A. Count concrete poured only

B. Track losses, exposure, maintenance and downstream impacts

C. Ignore land-use change

D. Attribute every flood solely to global climate change

Answer: B. Flood resilience depends on hazard and exposure as well as drainage, wetlands and governance; outcome indicators and spillovers are essential.

MCQ 29. Why is methane reduction useful alongside CO2 reduction?

A. Methane is not a greenhouse gas

B. Methane has important near-term warming effects

C. Methane stays forever in the atmosphere

D. CO2 reduction is unnecessary

Answer: B. Methane is a potent, shorter-lived greenhouse gas; reducing it can slow near-term warming, while sustained CO2 cuts are necessary for long-term stabilisation.

MCQ 30. Which statement is most accurate about climate scenarios?

A. They predict a single inevitable future

B. They explore outcomes under alternative assumptions

C. They eliminate local uncertainty

D. They are interchangeable with daily weather forecasts

Answer: B. Scenarios condition projections on future emissions, socioeconomic choices and model assumptions; they are tools for risk planning, not deterministic forecasts.

Six complete mains model answers (original practice)

Model answer 1. Explain the scientific basis of global warming and why one hot year does not define the Paris temperature threshold.

Background: Global warming means a persistent rise in global mean surface temperature, while climate change includes shifts in rainfall, extremes and other climatic characteristics. The distinction is essential for interpreting climate statistics and policy commitments.

Physical mechanism: Solar energy enters the Earth system, and outgoing infrared radiation is partly absorbed and re-emitted by greenhouse gases. Human emissions of carbon dioxide, methane and nitrous oxide strengthen this natural effect. Ocean heat uptake delays some surface response, while ice–albedo and water-vapour feedbacks amplify warming.

Evidence: The IPCC's 2023 synthesis finds human causation unequivocal and estimates 1.1°C warming for 2011–2020 relative to 1850–1900. WMO's March 2026 assessment placed 2025 near 1.43°C above the same baseline, with 2015–2025 the eleven warmest years recorded. Independent ocean, ice and atmospheric observations corroborate the trend.

Interpretation: Annual temperature varies with El Niño, volcanic effects and other factors. A single calendar year above 1.5°C is therefore not equivalent to sustained long-term warming beyond the Paris objective. Different observational datasets can also produce slightly different annual rankings.

Conclusion: Sound policy also requires local participation, measurable implementation, and transparent public accountability. Climate policy should rely on long-term observations, attribution science and transparent baselines, not

anecdotes about individual seasons. Rapid emissions cuts and risk-informed adaptation are complementary responses.

Model-answer length: 207 words.

Model answer 2. Assess the climate-change risks for Telangana agriculture and recommend a practical adaptation strategy.

Context: Telangana combines rainfed farming, irrigated crops, tank systems and borewell-dependent agriculture. Climate change can intensify heat stress, rainfall variability and the risk of damaging extremes, although local land and water management strongly influence outcomes.

Risk pathways: Delayed monsoon onset can disturb sowing, high temperature during flowering can reduce yields, and concentrated rainfall may erode soil or damage harvests. Repeated dry spells increase irrigation demand, while groundwater depletion magnifies vulnerability. Crop productivity and farm income are not identical because prices, debt and market access matter.

Adaptation package: District agriculture departments should integrate agrometeorological advisories, resilient seed varieties, crop diversification, soil-moisture conservation and restoration of appropriate tank and watershed systems. Groundwater budgets must accompany irrigation expansion. Farmers shifting away from water-intensive crops need procurement, storage, extension and price-risk protection.

Inclusion and implementation: Tenant farmers, smallholders and women cultivators must be eligible for advisories, insurance and credit. Agricultural universities, local institutions and EPTRI can support climate-risk assessment. Projects should compare outcomes against baseline conditions rather than count only subsidy beneficiaries.

Conclusion: Sound policy also requires local participation, measurable implementation, and transparent public accountability. A successful strategy protects net farm income and water security together. Evaluation should track yield stability, groundwater levels, input costs, household debt and access for vulnerable cultivators over several seasons.

Model-answer length: 211 words.

Model answer 3. Compare mitigation and adaptation with examples from Hyderabad and Telangana.

Definitions: Mitigation reduces greenhouse-gas emissions or enhances credible removals; adaptation adjusts systems to actual or expected climatic effects. Both are essential because past emissions already create risks while future emissions determine the severity of longer-term warming.

Mitigation measures: Hyderabad can expand dependable public transport, efficient buildings, distributed solar energy, waste-methane control and cleaner industrial processes. These reduce emissions when their lifecycle and electricity-system effects are properly counted. Renewable installed capacity alone should not be confused with actual clean electricity generation.

Adaptation measures: Heat alerts, shaded public spaces, cool roofs, reliable drinking water, restored urban lakes and maintained drainage can reduce exposure. Rural Telangana needs drought planning, water budgeting, crop advisories and resilient public health services. These measures often deliver immediate welfare benefits without materially changing global emissions.

Synergies and risks: Urban trees can moderate heat and store some carbon, but survival and water requirements matter. Air-conditioning can save lives while increasing peak power demand; energy-efficient passive cooling reduces that tension. Unrestricted groundwater pumping during drought may create maladaptation by depleting aquifers.

Governance: Municipalities, state departments, EPTRI and community organisations should use measurable indicators: avoided heat illness, flood-loss reduction, reliable power, emissions per service and inclusion of informal workers.

Conclusion: Climate resilience requires a balanced portfolio. Neither mitigation without protection of vulnerable people nor adaptation without emissions reduction can deliver sustainable development.

Model-answer length: 221 words.

Model answer 4. Examine India’s 2031–2035 NDC and distinguish commitments from observed achievements.

Context: Nationally determined contributions are countries' climate commitments under the Paris Agreement. India submitted its 2031–2035 NDC to the UNFCCC on 24 April 2026, linking climate action to development, equity and international cooperation.

Quantified commitments: The document targets a 47% reduction in GDP emissions intensity by 2035 relative to 2005; about 60% cumulative installed electric-power capacity from non-fossil sources by 2035; and a forest-and-tree-cover carbon sink of 3.5–4.0 billion tonnes of CO2 equivalent by 2035 compared with 2005. It also stresses adaptation, Mission LiFE, finance and technology.

Measurement distinctions: Emissions intensity is emissions per unit GDP, not absolute emissions. Installed capacity is not electricity actually generated. Carbon-sink targets require credible accounting and permanence. The NDC explicitly avoids imposing individual-sector reduction obligations, including on agriculture.

Observed position: The same official document reports that non-fossil sources accounted for 52.57% of installed electric-power capacity on 28 February 2026. This is a dated observed statistic, not proof that the 2035 target has been fulfilled. India’s net-zero goal for 2070 is a separate long-term objective.

Conclusion: Sound policy also requires local participation, measurable implementation, and transparent public accountability. Implementation requires reliable grids, finance, technology, ecological safeguards and a just transition. Examination answers must label baselines, years, targets and outcomes precisely.

Model-answer length: 205 words.

Model answer 5. Discuss sustainable development and the policy tensions between growth, equity and ecology.

Meaning: Sustainable development meets present needs without undermining the ability of future generations to meet theirs. It integrates economic opportunity, social justice and ecological integrity rather than treating environmental protection as an optional addition to GDP growth.

Principles: Intergenerational equity protects future people; intragenerational equity addresses unequal burdens today. The precautionary principle supports action against serious possible harm despite uncertainty. Polluter-pays encourages responsible parties to bear pollution-control and damage costs. Participation and transparency improve legitimacy.

Trade-offs: A dam may support water supply but disrupt downstream ecosystems. Renewable-energy parks reduce fossil dependence but can compete with grazing or agricultural land. An industrial estate may generate jobs while stressing local groundwater. Policy should compare alternatives, compensate affected groups, maintain critical ecological services and measure lifecycle effects.

Institutions and indicators: Environmental impact assessment, public consultation, urban planning, water budgeting and independent monitoring complement financial appraisal. The UN 2030 Agenda's 17 goals and 169 targets link climate action with health, energy, cities, water, biodiversity and reduced inequality. Telangana can use disaggregated indicators for flood risk, air quality, farm incomes and water security.

Conclusion: Sustainable development is a governance discipline, not a slogan. Projects should be judged by who benefits, who bears costs, whether ecological limits are respected and whether gains endure across generations.

Model-answer length: 209 words.

Model answer 6. Evaluate international climate governance and the importance of climate justice for developing countries.

Institutional evolution: The UNFCCC of 1992 established a cooperative framework and the principle of common but differentiated responsibilities and respective capabilities. The Kyoto Protocol assigned quantified commitments to certain developed countries. The Paris Agreement of 2015 uses nationally determined contributions and a transparency framework involving all parties.

Paris architecture: Article 2 aims to keep warming well below 2°C and pursue efforts toward 1.5°C, strengthen adaptation and align finance with climate-resilient development. Periodic NDC updates and global stocktakes assess collective ambition. Targets, COP decisions and actual emissions outcomes must be distinguished.

Justice concerns: Developed countries have contributed disproportionately to historical cumulative emissions and generally possess greater financial and technological capacity. Developing countries emphasise poverty eradication, affordable energy, adaptation needs and fair access to the remaining carbon budget. Equity cannot remove the physical need to cut emissions, but it influences fair burden-sharing.

Implementation barriers: Finance pledges may not translate into accessible grants; debt-heavy loans can burden vulnerable states. Technology transfer requires skills, maintenance and intellectual-property arrangements. Loss and damage also includes non-economic harms that conventional insurance may not capture.

Conclusion: Effective governance combines stronger mitigation with predictable adaptation finance, transparent reporting and just transitions. India's approach links domestic low-carbon development with CBDR-RC, while states such as Telangana translate global objectives into local water, health and agricultural resilience.

Model-answer length: 215 words.

Dated primary-source register and audit

- **S01 | IPCC | 2023.** AR6 Synthesis Report headline statements. https://report.ipcc.ch/ar6syr/headline.html — Used for: Human attribution, 2011–2020 warming, impacts and adaptation.

- **S02 | WMO | 23 March 2026.** State of the Global Climate 2025. https://wmo.int/publication-series/state-of-global-climate/state-of-global-climate-2025 — Used for: 2025 anomaly, ocean heat, 2015–2025 ranking.

- **S03 | Copernicus C3S | 14 January 2026.** 2025 third warmest year. https://climate.copernicus.eu/copernicus-2025-was-third-hottest-year-record — Used for: Dataset ranking and short-term exceedance caveat.

- **S04 | UNFCCC | April 2026; submitted 24 April 2026.** India NDC (2031–2035), official PDF. https://www.unfccc.int/sites/default/files/2026-04/INDIA%20NDC%202031-35.pdf — Used for: 2035 commitments and observed February 2026 installed capacity.

- **S05 | UNFCCC | 24 April 2026.** India NDC submission registry. https://unfccc.int/documents/611410 — Used for: Submission date.

- **S06 | UN DESA | September 2015.** Transforming our world: 2030 Agenda. https://sdgs.un.org/2030agenda — Used for: 17 SDGs, 169 targets, integration and equity.

- **S07 | UN DESA | accessed October 2026.** Sustainable Development Goals overview. https://sdgs.un.org/goals — Used for: Goal mapping and institutional context.

- **S08 | Telangana EPTRI | 2015 plan; 2016 GO; current web record.** Climate Change Centre and TSAPCC. https://www.eptri.telangana.gov.in/climate — Used for: EPTRI nodal status and reported state initiatives.

- **S09 | UNFCCC | 2015.** Paris Agreement. https://unfccc.int/process-and-meetings/the-paris-agreement — Used for: Article 2 temperature and finance goals.

- **S10 | UNFCCC | 1992.** Convention text and principles. https://unfccc.int/resource/docs/convkp/conveng.pdf — Used for: CBDR-RC framework.

- **S11 | IPCC | 2022.** AR6 Working Group II Impacts, Adaptation and Vulnerability. https://www.ipcc.ch/report/ar6/wg2/ — Used for: Hazards, vulnerability and adaptation.

- **S12 | IPCC | 2022.** AR6 Working Group III Mitigation. https://www.ipcc.ch/report/ar6/wg3/ — Used for: Energy and mitigation pathways.

- **S13 | Government of India | 2008.** National Action Plan on Climate Change overview. https://moef.gov.in/ — Used for: NAPCC historical institutional context; verify mission count by year.

- **S14 | United Nations | 1987.** Brundtland Report Our Common Future. https://sustainabledevelopment.un.org/content/documents/5987our-common-future.pdf — Used for: Sustainable development definition.

- **S15 | UNDRR | 2015–2030.** Sendai Framework for Disaster Risk Reduction. https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030 — Used for: Risk governance.

- **S16 | UNFCCC | 2023.** First Global Stocktake COP28 outcome. https://unfccc.int/topics/global-stocktake — Used for: Collective progress and energy transition.

- **S17 | UNEP | 2025.** Emissions Gap Report. https://www.unep.org/resources/emissions-gap-report-2025 — Used for: Emissions gap contextual source; no unaudited numerical claim.

- **S18 | WHO | current.** Climate change and health fact sheet. https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health — Used for: Heat and health pathways.

- **S19 | India Meteorological Department | current.** Weather and climate services. https://mausam.imd.gov.in/ — Used for: Climate and warning institutions.

- **S20 | Telangana Government | current.** State portal and government institutions. https://www.telangana.gov.in/ — Used for: Telangana governance context.

- **S21 | UNFCCC | 1997.** Kyoto Protocol. https://unfccc.int/kyoto_protocol — Used for: Historical climate treaty distinction.

- **S22 | UNEP | 2016 amendment.** Montreal Protocol and Kigali Amendment. https://ozone.unep.org/treaties/montreal-protocol — Used for: Ozone versus climate treaty distinction.

Source and accuracy audit

- **Directly checked official 2026 data:** WMO State of Global Climate 2025; UNFCCC India NDC April 2026 and submission date; UN 2030 Agenda; EPTRI institutional information available in its public record.

- **Syllabus limitation:** The exact Topic 5 wording is corroborated by a public reproduction of Notification 02/2024; a direct commission-hosted PDF could not be retrieved during this run. Academic sign-off requires checking the original PDF.

- **Caution on current figures:** Annual temperature anomalies are not long-term Paris thresholds. Capacity is not generation. Intensity is not absolute emissions. All 2035 NDC numbers are targets; the February 2026 capacity share is dated observed evidence.

- **Institutional caveat:** EPTRI's project descriptions are administrative claims, not independent outcome evaluations. NAPCC mission counts differ in later official descriptions; specify original eight and verify later revisions.

- **Question provenance:** All questions in this chapter are original; no unverified official PYQs have been invented.

- **Academic status:** NOT APPROVED. Independent syllabus, legal, data, language and examination review remains outstanding.